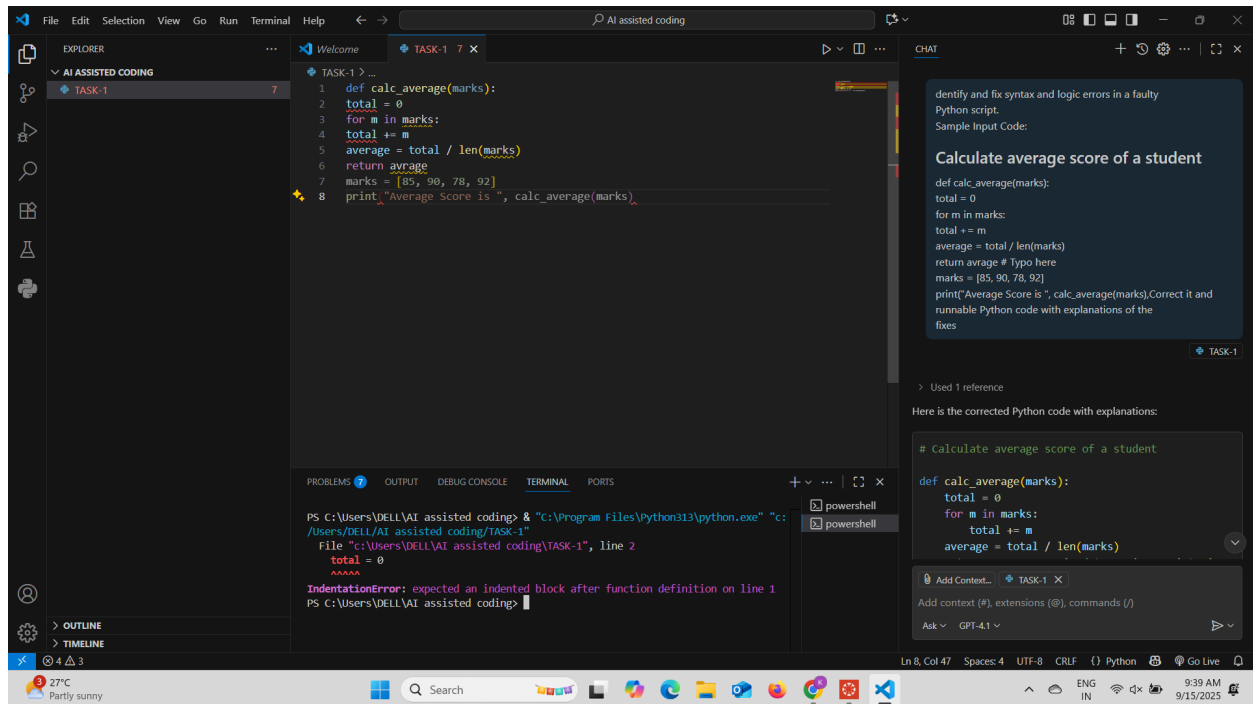
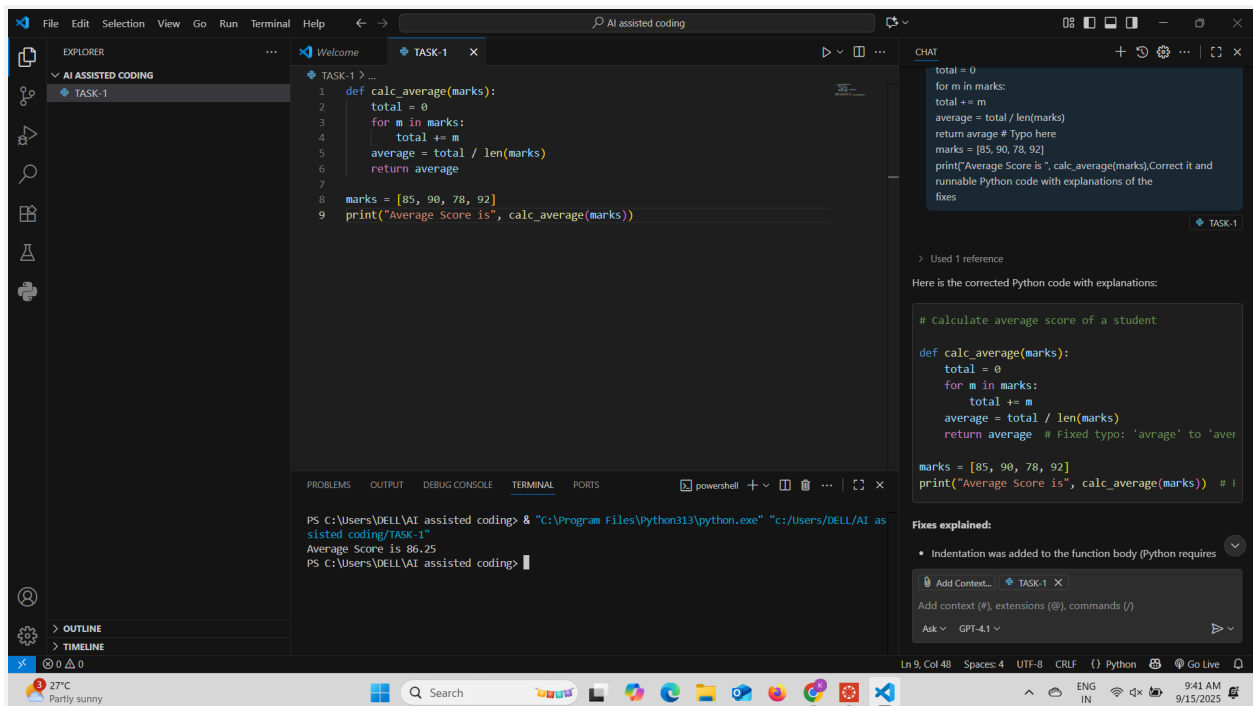


Roll Number :2403A51457  
AI ASSISTED CODING,Assignment(10.1)

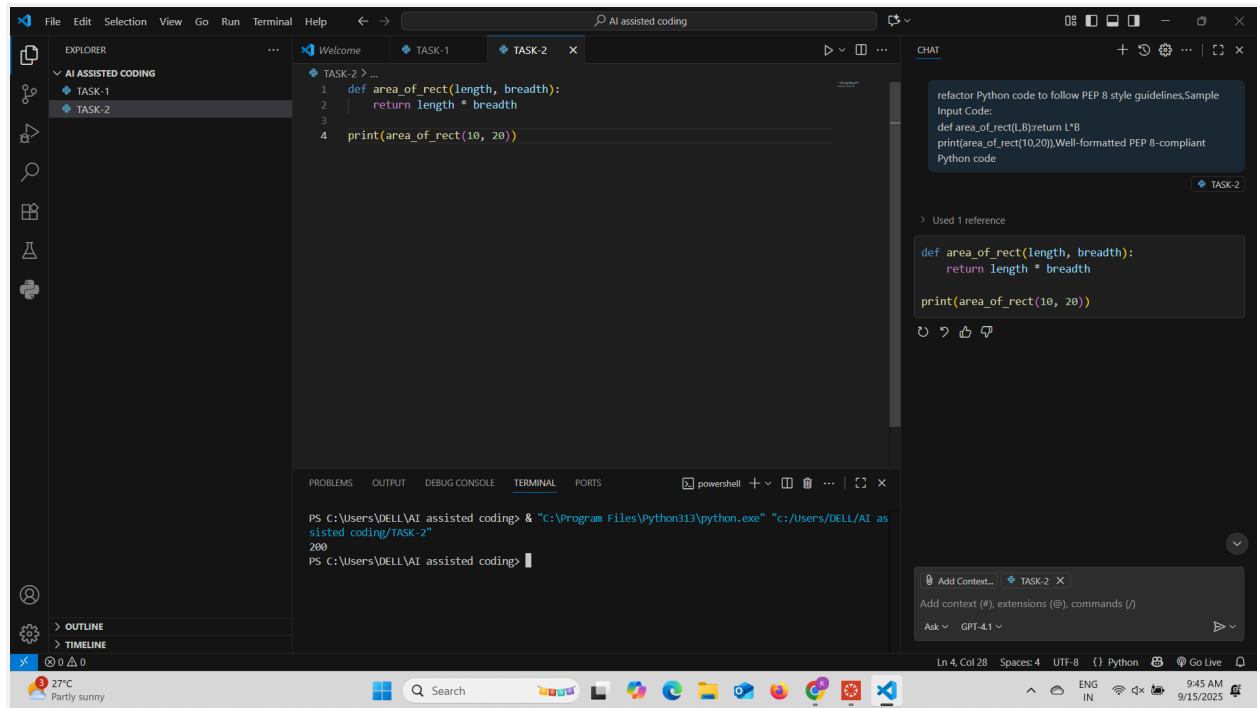
## TASK-1:



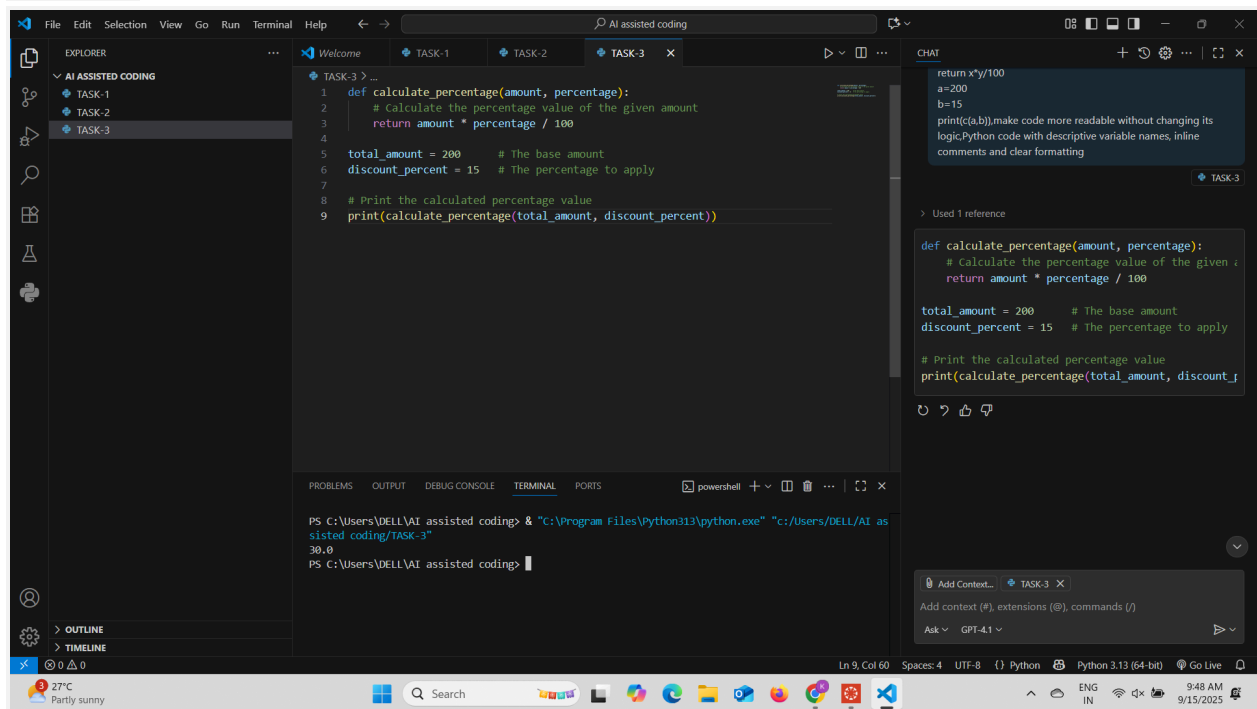
## Fixed syntax and logic errors:



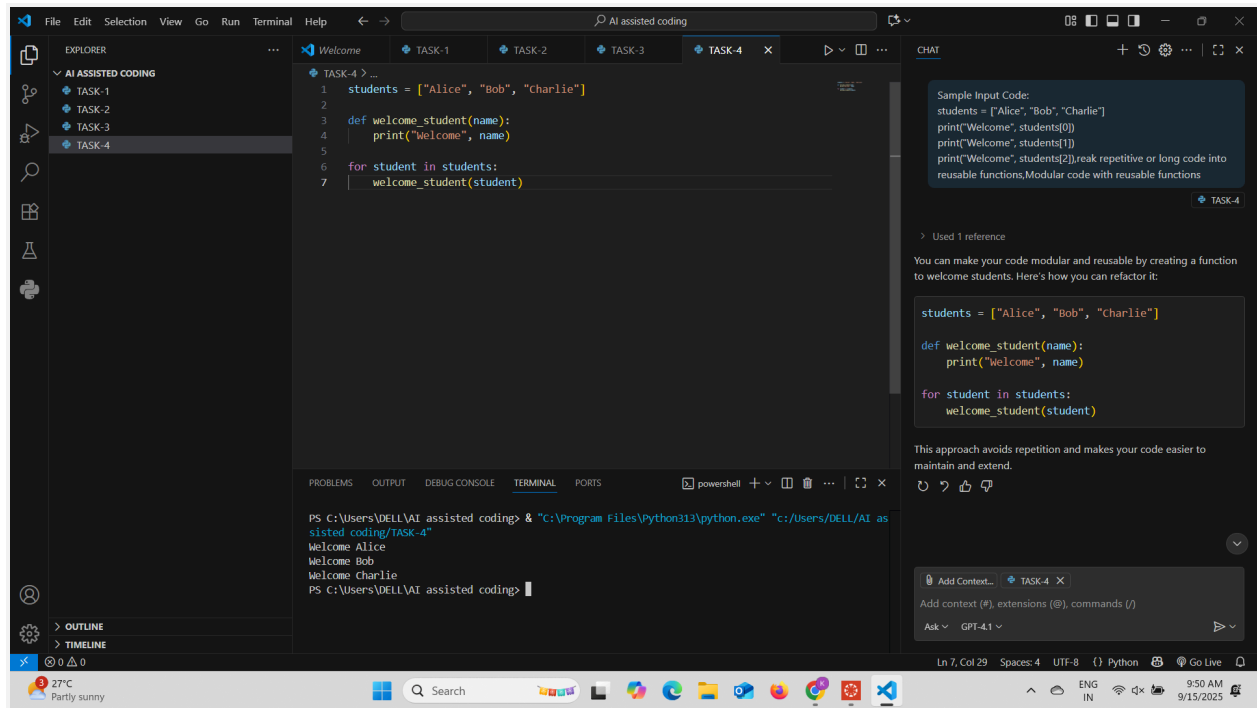
## TASK-2:



## TASK-3:

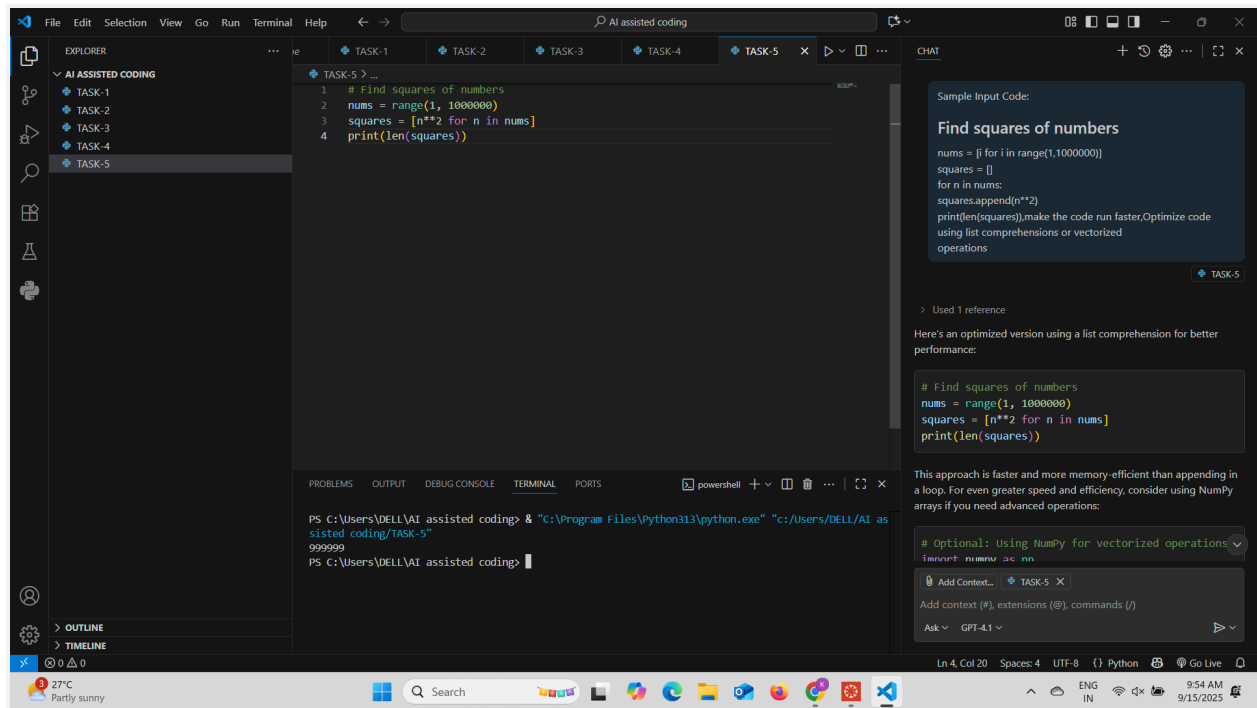


## TASK-4:



## TASK-5:

### # finding Squares of numbers:



## TASK-6:

#Here's a cleaner version using elif statements:

The screenshot shows the VS Code interface with a Python file named 'TASK-6'. The code defines a 'grade' function that takes a score and returns a letter grade based on ranges. The current code uses multiple 'if' and 'elif' statements. The AI chat panel on the right provides two alternative suggestions: one using 'elif' statements and another using a dictionary mapping with ranges. The terminal at the bottom shows the execution of the script, outputting 'B', 'C', and 'F' for scores 80, 70, and 59 respectively.

```
def grade(score):
    if score >= 90:
        return "A"
    elif score >= 80:
        return "B"
    elif score >= 70:
        return "C"
    elif score >= 60:
        return "D"
    else:
        return "F"

print(grade(80))
print(grade(70))
print(grade(59))
```

Here's a cleaner version using `elif` statements:

```
def grade(score):
    if score >= 90:
        return "A"
    elif score >= 80:
        return "B"
    elif score >= 70:
        return "C"
    elif score >= 60:
        return "D"
    else:
        return "F"
```

Alternatively, using a dictionary mapping with ranges:

```
def grade(score):
    grade_map = {
        (90, 100): "A",
        (80, 89): "B",
        (70, 79): "C",
        (60, 69): "D",
        (0, 59): "F"
    }
    for (low, high), letter in grade_map.items():
        if low <= score <= high:
            return letter
```

```
PS C:\Users\DELL\AI assisted coding> & "C:\Program Files\Python313\python.exe" "C:\Users\DELL\AI assisted coding\TASK-6"
B
C
F
PS C:\Users\DELL\AI assisted coding>
```

#Using a dictionary mapping with ranges:

This screenshot is similar to the previous one, but the code in the editor already uses a dictionary mapping for the grade function. The AI chat panel on the right provides a suggestion for a cleaner version using a dictionary mapping with ranges, which is identical to the code already in the editor. The terminal at the bottom shows the execution of the script, outputting 'B', 'C', and 'F' for scores 80, 70, and 59 respectively.

```
def grade(score):
    grade_map = {
        (90, 100): "A",
        (80, 89): "B",
        (70, 79): "C",
        (60, 69): "D",
        (0, 59): "F"
    }
    for (low, high), letter in grade_map.items():
        if low <= score <= high:
            return letter

print(grade(80))
print(grade(70))
print(grade(59))
```

Alternatively, using a dictionary mapping with ranges:

```
def grade(score):
    grade_map = {
        (90, 100): "A",
        (80, 89): "B",
        (70, 79): "C",
        (60, 69): "D",
        (0, 59): "F"
    }
    for (low, high), letter in grade_map.items():
        if low <= score <= high:
            return letter
```

```
PS C:\Users\DELL\AI assisted coding> & "C:\Program Files\Python313\python.exe" "C:\Users\DELL\AI assisted coding\TASK-6"
B
C
F
PS C:\Users\DELL\AI assisted coding>
```